

Effective Visuals Design: Key Concepts

This lesson focuses on how to design effective visuals by managing cognitive load and reducing clutter. The main goal is to present information in a way that is easy for the audience to understand and process.

Cognitive Load

Cognitive load refers to the mental effort required to learn new information. When presenting data, you are using the audience's mental processing power.

- **Goal:** The aim is to make **optimal use** of this cognitive load to deliver information effectively.
 - **Problem:** Unnecessary visual elements, known as **clutter**, increase cognitive load and make it harder for the audience to understand the main message.
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Clutter in Data Visualization

Clutter consists of visual elements that take up space but do not add to the understanding of the information. It complicates the visual and hinders effective communication.

Categories and Types of Clutter

Main Categories of Clutter

Visuals can be broken down into five main components where clutter can occur:

1. **Frame:** The background and borders of the chart.
2. **Graphical objects:** The bars, lines, or points used to represent data.
3. **Axes:** The reference lines for measurement.
4. **Labels:** All text elements, including titles, axis labels, and legends.
5. **Overlay objects:** Explanatory elements like callout boxes that are placed over the chart.

Other Common Types of Clutter

- **Lack of Visual Order:** When elements are not logically or neatly arranged, the visual can feel chaotic. Strategic use of color and alignment can bring order. For example, in a bar chart showing survey feedback, highlighting the most important result with a dark color and making other bars a muted grey creates a clear focal point.
- **Poor Alignment:** Text and graphical elements should be aligned to create a clean, professional look. Avoid center-aligning text; left- or right-aligning text is typically easier to read.

- **Ineffective Use of White Space:** White space is the empty area around elements. Just like pauses in a speech, strategic use of white space can draw attention to the important parts of your visual.
 - **Non-strategic Use of Contrast:** Contrast helps direct your audience's attention. However, if you try to make too many things stand out (e.g., using many different colors and shapes in a scatter plot), nothing will stand out. A better approach is to use a single color, like blue, for your data and grey for competitors to make your performance easy to see.
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Step-by-Step Guide to Decluttering a Chart

Let's use an example of a line graph showing "Ticket Volume Received" versus "Ticket Volume Processed" over a year to illustrate how to declutter.

The Goal: Simplify the visual to clearly show the relationship between tickets received and processed, especially during months when team members left, to help decide if new hires are needed.

1. **Remove the Chart Border:** A chart border is usually unnecessary and adds visual noise. The human brain automatically sees the chart elements as a cohesive whole (this is the Gestalt principle of closure). White space can be used to separate the chart from other page elements.
2. **Remove Gridlines:** Gridlines can also be distracting and are often not needed. If some guidance is necessary to read the values, make the gridlines a very light, muted color, like a faint grey, so they don't compete with the data.
3. **Remove Data Markers:** Data markers (the dots or squares on a line graph) are often redundant because the lines already show the data's path. Only include them if they serve a specific purpose.
4. **Clean Up Axis Labels:**
 - **Y-Axis (Vertical):** Remove unnecessary precision. For example, if the values are whole numbers, change labels like "150.00" to "150".
 - **X-Axis (Horizontal):** Abbreviate labels for readability. Change "January," "February," etc., to "Jan," "Feb" to avoid diagonal text, which is hard to read.
5. **Label Data Directly:** Instead of using a separate legend, place the labels ("Received" and "Processed") directly next to the corresponding lines. This makes it easier for the audience, as they don't have to look back and forth between the legend and the data.
6. **Use Consistent Color:** Match the color of the text label to the color of the data line it describes. For example, make the "Received" line and its label blue, and the "Processed" line and its label red. This reinforces the connection between the labels and the data.